

COMP4213 Wireless Communications and Mobile Computing: Final Project (30%)

1 Problem Description

The final project is to develop a mobile application called “BNBU-Nav” that can localize oneself in a building at BNBU. The building should have at least three floors, each floor with at least three rooms. You can use wireless signals, such as WiFi, Bluetooth, with their signal strengths or channel state information. You can also leverage other sensor signals, including cameras, IMUs, etc.

2 Basic Features (20%)

Your application will be evaluated based on the following criteria.

1. Identify the person on the correct floor (10%).
2. Identify the person in the correct room (10%).

3 Advanced Features (10%)

Other 10% will be given to extra features implemented, such as

1. Design a path from a room to another room at BNBU. The destination room should be at least one floor apart (2%).
2. Navigate the person to the destination. The destination room should be at least one floor apart (5%).
3. The APP works between multiple buildings (5%).
4. Display real-time or future (in 5 days) school events on the map by crawling information from school website (5%).
5. Enable augmented reality features during navigation (5%).
6. Other interesting features that you can think of.

4 Presentation

A group presentation of the APP shall be delivered at the end of the semester in class, from December 9 to December 16. **EVERY TEAM MEMBER** should present. A short demo video (30 seconds to 1 minute) is required during the presentation. The length and the order of the presentations will be announced later when the groups are formed.

5 Other Requirements

1. Group size: 3 – 5.
2. Programming language: Any.
3. Mobile platform: Any.

6 Things to submit

1. Runnable code for the application with detailed documentation.
2. A summary report of the application and the contribution (in percentage) of each team member. Remember to put your group number as well as the names and student IDs of all the team members.
3. The presentation PPT and the demo video.